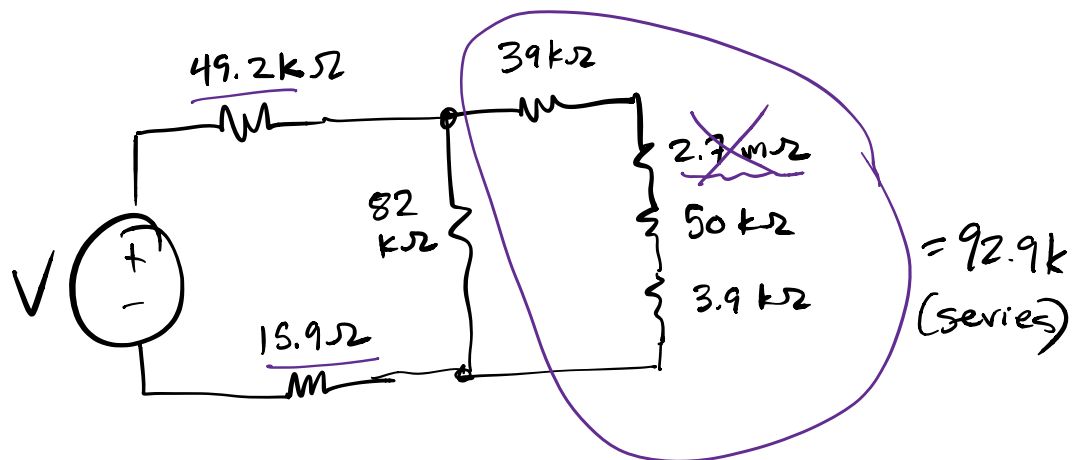
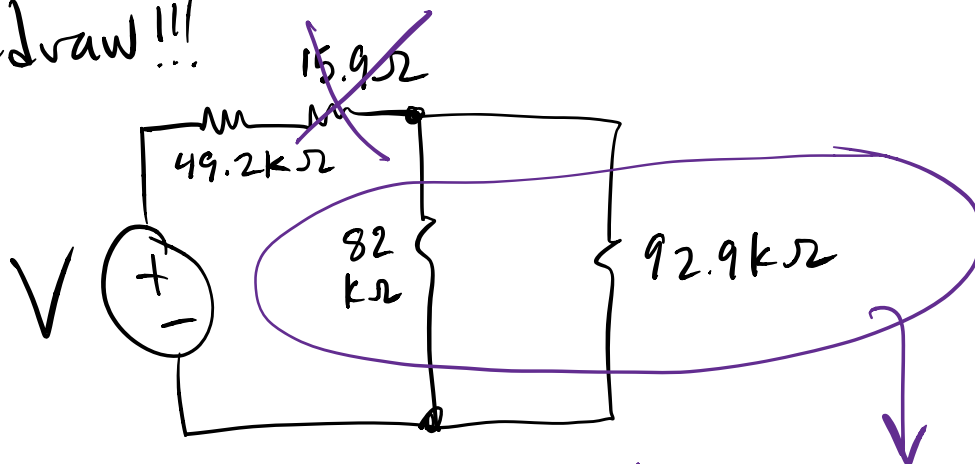




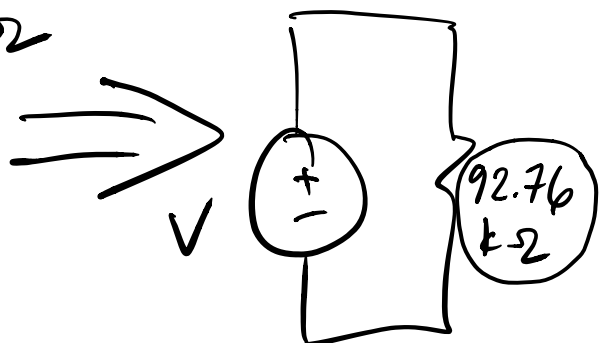
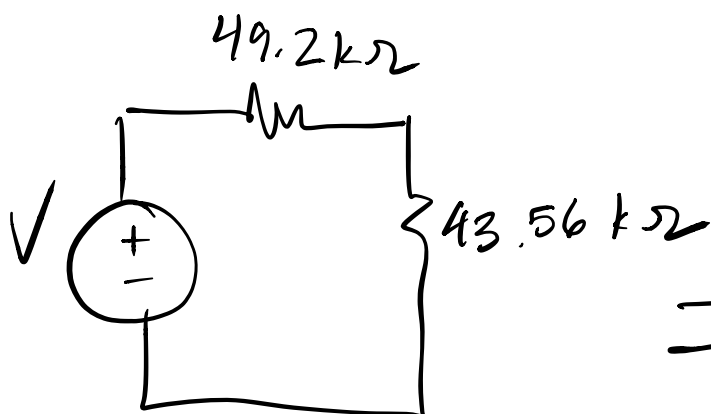
$2.7\text{ m}\Omega \ll \ll \ll \ll \text{ k}\Omega$

redraw!!!

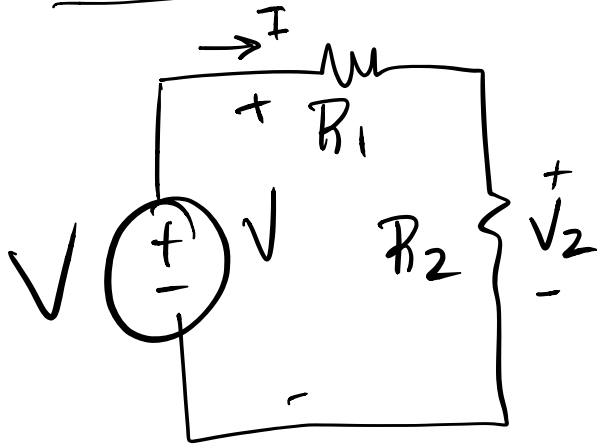


$15.9\text{ }\Omega \ll \ll 49.2\text{ k}\Omega$

$$\frac{82 \cdot 92.9}{82 + 92.9} = 43.56\text{ k}\Omega$$



Voltage Division



.. let's say you only want the voltage across R_2 , called V_2

.. if you don't care what the current is, or any other voltages, why do a KVL loop?

.. we know
$$I = \frac{V}{R_1 + R_2}$$

.. we also know
$$V_2 = I R_2$$

∴
$$V_2 = V \left(\frac{R_2}{R_1 + R_2} \right) = \frac{V}{R_1 + R_2} \cdot R_2$$

Universal voltage divider equation :

$$\frac{V_x}{V} = \frac{R_x}{R_{eq.}}$$

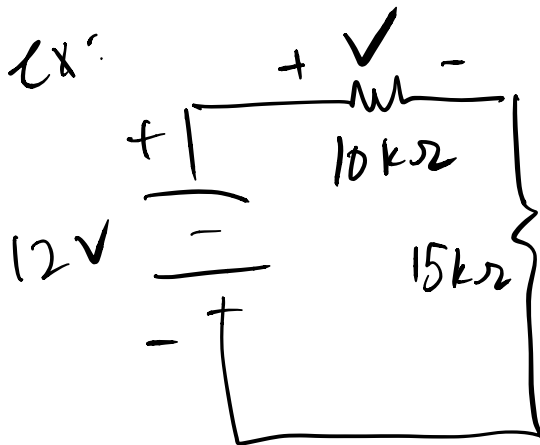
the voltage you want across its resistance

∴ $V_x = V \left(\frac{R_x}{R_{eq.}} \right)$

numerator is always whatever your desired voltage is!

-- always a fraction of original source voltage; $V_x \leq V$

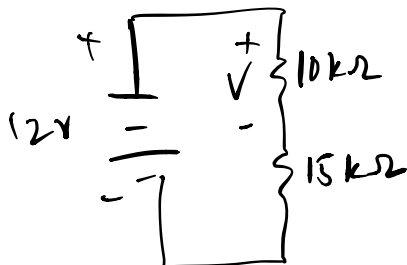
-- if you wind up with more voltage than you started with, you're in trouble!



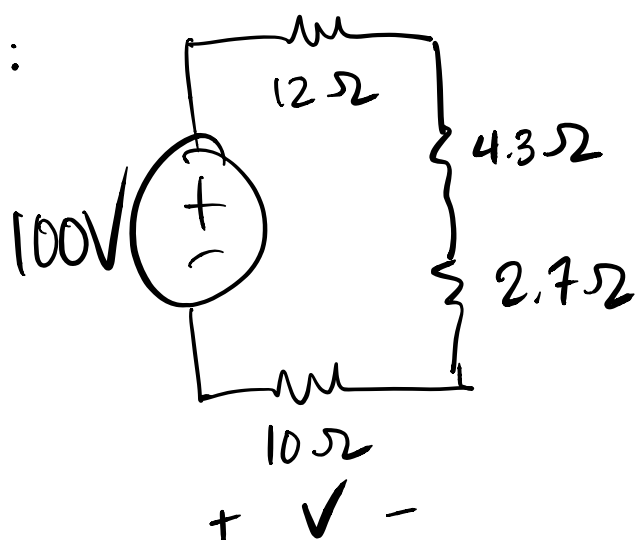
-- determine V

$$V = 12 \left(\frac{10k}{10k + 15k} \right)$$

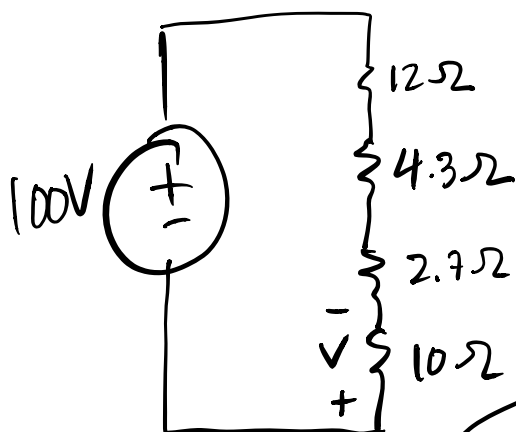
$$V = 4.8V$$



ex:



~ determine V



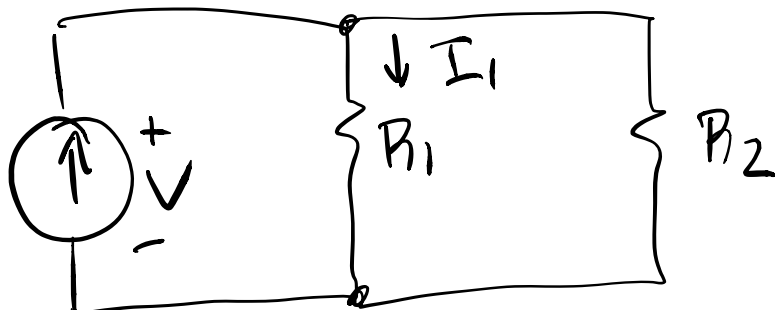
.. note polarity of V !!!

$$-V = 100 \left(\frac{10\Omega}{12\Omega + 4.3\Omega + 2.7\Omega + 10\Omega} \right)$$

$$-V = 34.48 \text{ V}$$

$$V = -34.48 \text{ V}$$

Current Division



.. let's say we just want I_1 and don't care about any other currents or voltages

.. we know $V = I \left(\frac{R_1 R_2}{R_1 + R_2} \right)$

parallel combination of R_1 and R_2

.. we also know $I_1 = \frac{V}{R_1}$

so $I_1 = I \frac{\cancel{R_1} R_2 / (R_1 + R_2)}{\cancel{R_1}}$

$I_1 = I \frac{R_2}{R_1 + R_2}$

looks like voltage divider equation, except R_2 is the other resistor!

.. this idea doesn't work for more than two resistors!

.. need to back up one step, don't cancel R_1 !

current divider equation:

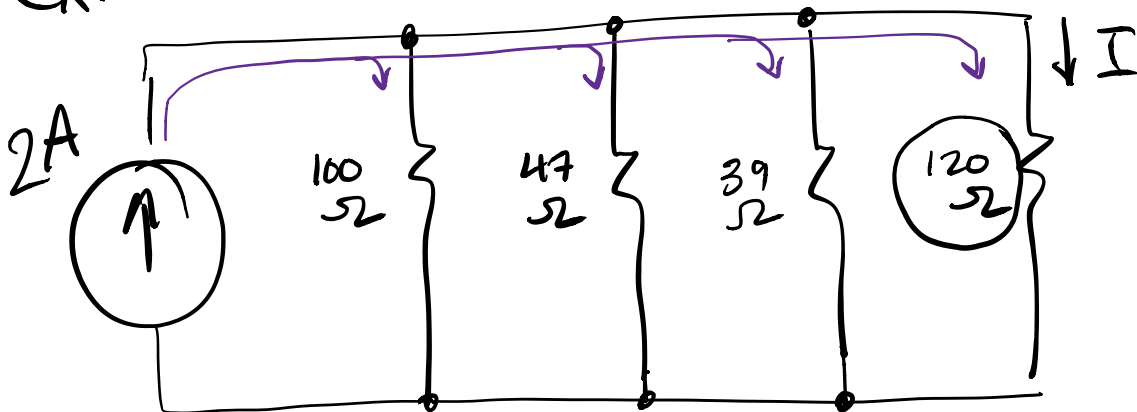
$$\frac{I_x}{I} = \frac{R_{eq.}}{R_x}$$

.. current you want

.. note that a parallel combination always has less resistance than any one resistor

∴ if you wind up with more current than you started with: run!

ex:



.. in this example, I will have the same sign as the driving current source

.. watch signs, redraw if necessary!

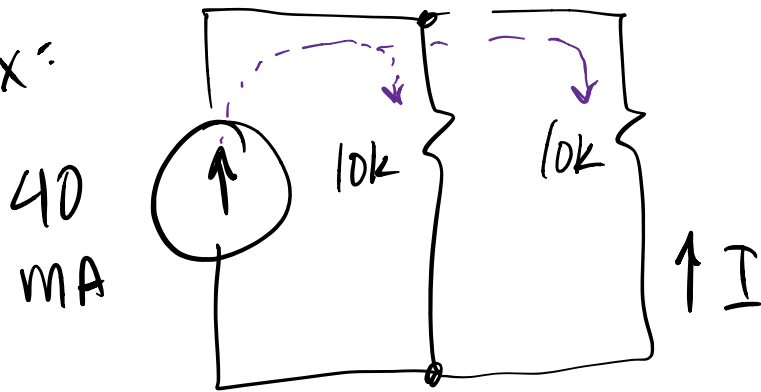
.. find Req. first

4

$$R_{eq.} = \frac{1}{\frac{1}{100} + \frac{1}{47} + \frac{1}{39} + \frac{1}{120}} = \underline{15.33 \Omega}$$

$$I = 2 \left(\frac{15.33}{120} \right) = \underline{0.2554 A}$$

ex:



.. note direction of I !

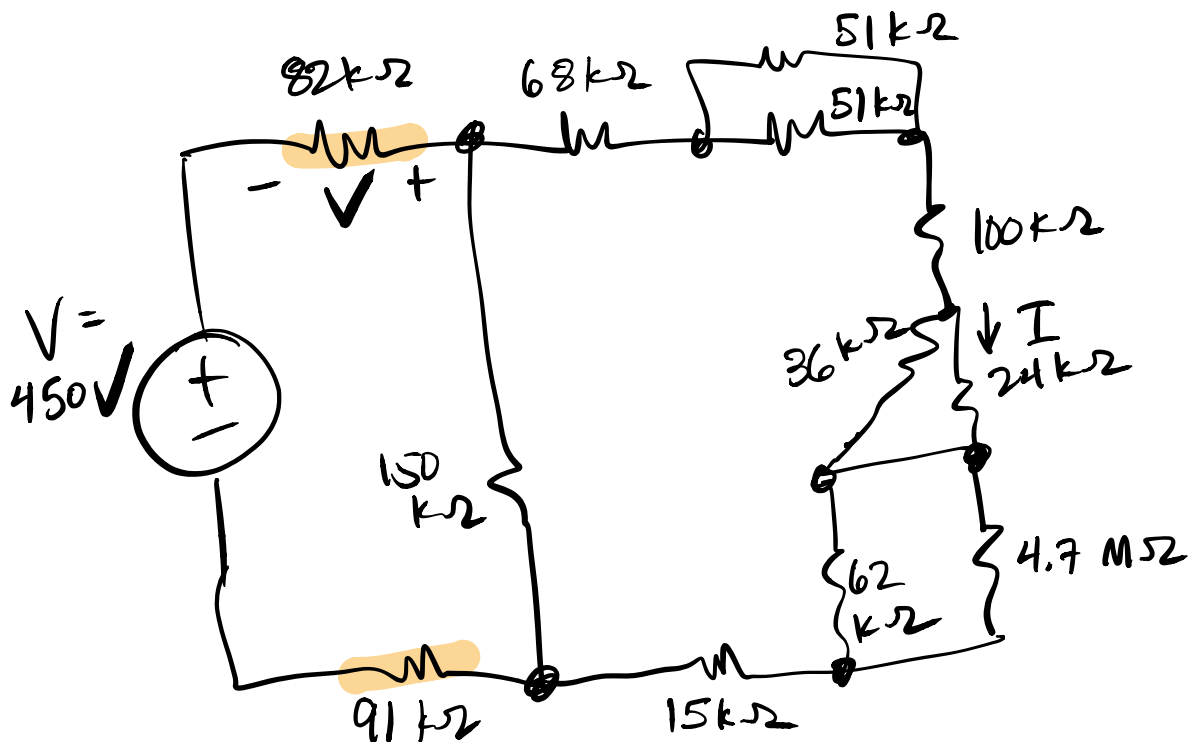
$$-I = 40 \left(\frac{10k \parallel 10k}{10k} \right)$$

.. equal-value resistors in parallel $\rightarrow \frac{R}{n}$

$$-I = 40 \left(\frac{5k}{10k} \right) = 20 mA = \underline{5k}$$

$$\underline{I = -20 mA}$$

- .. voltage dividers w/ equal resistors divide voltage by n
- .. current dividers w/ equal resistors divide current by n
(but watch signs!)



- .. determine V and I using voltage;
current division
- .. when combining resistors, leave the ones
with the quantities of interest alone!